MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPES 1/ DOD-023, 2/ DOD-024, 3/ DOD-025

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Voltage tunable, low noise, integral magnet, 36 mW minimum power output, frequency range (see chart below).

ABSOLUTE RATINGS:

Parameter:	Ef	If	Eb	Ib	Ec	Ic	σ'	TA	Alt
Unit:	V	A	Vdc	mAdc	Vdc	mAdc	VSWR	°C	ft
Maximum:	3.0	2.1	1,700	20	250	1.0	2:1	100	100,000
Minimum:	1.3	1.9	900	1.0	50	0	---	---	

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Marking:	See figure 1 and note 26
Mounting position:	Any	Cathode:	Cold cathode (external connection required) - See figure 1
RF output impedance:	50 ohms		
Coupling:			
Input:	Flying leads 18 inches long, flexible silicon rubber insulated	Magnet isolation:	See note 25
		Weight:	4 pounds (maximum)
Output:	Type TNC (female)	Cooling:	Convection and conduction

TEST CONDITIONS:

Parameter:	Ef	If	Eb	Ib	Ec	Ic
Unit:	V	A	Vdc	mAdc	Vdc	mAdc
Maximum:	3.0	2.1	Set up from data on tube label and individual test data sheet (see note 26)			
Minimum:	1.3	1.9				

	FREQUENCY (MHz)		
F	Tube type 1/ DOD-023	Tube type 2/ DOD-024	Tube type 3/ DOD-025
1	885	1,420	2,507
2	1,035	1,717	2,958
3	1,185	2,014	3,409
4	1,335	2,311	3,860
5	1,485	2,607	4,310

GENERAL:

Qualification - Not required

1/ Replaces Cutler-Hammer Part No. 146002-2
General Electric Type No. ZM-6085
2/ Replaces Cutler-Hammer Part No. 146002-1
General Electric Type No. ZM-6086
3/ Replaces Cutler-Hammer Part No. 146003-1
General Electric Type No. ZM-6087

DOD-023, DOD-024, DOD-025

FSC 5960

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1	1, 2					
1027	Temperature cycling, nonoperating	3	3 cycles	---	---	---	---
1031	Nonoperation vibration	4	No voltages	---	---	---	---
4302	Frequency change	5	F1 thru F5	t	---	15	sec
4289	Heater current, nonoperating	-	Ef = 1.3 to 3.0 V	If	1.9	2.1	A
1261	Electrode voltage (injection)	-		Ec	50	250	Vdc
1256	Electrode current (injection)	-		Ic	0	1.0	mAdc
4280	Electronic tuning range	6, 9					
	1/ DOD-023	-		F	885	1,485	MHz
	2/ DOD-024	-		F	1,420	2,607	MHz
	3/ DOD-025	-		F	2,507	4,310	MHz
---	Linearity (tuning curve)	7	σ' = 1.2:1 (worst phase)	---	---	±2	%
---	Slope deviation (tuning curve)	8	σ' = 1.2:1 (worst phase)	---	---	±15	%
---	Initial accuracy (tuning curve)	9		---	---	±2.5	%
4310	Frequency pulling figure	10		ΔF	---	±0.15	%
4250	Power output	11	σ' = 1.2:1 (worst phase); F1 thru F5	Po	36	---	mW
4251	Output power variation	12		ΔPo	---	6	dB
4264	Ion noise modulation	13		Pn	85	---	dB
	Quality conformance inspection, part 2	2					
---	Electronic tuning repeatability	14	σ' = 1.2:1 (min)	---	---	±0.1	%
4030	Thermal frequency drift	15	T = -18° to +74° C	---	---	±0.2	%
4308	RF bandwidth	16		BW	---	3.0/tpc	MHz
4243	Spurious output ratio	17		spo	60	---	dB
4274	Voltage tuning characteristics	18					
	1/ DOD-023	-		F/Eb	1.0	---	MHz/v
	2/ DOD-024	-		F/Eb	1.66	---	MHz/v
	3/ DOD-025	-		F/Eb	2.88	---	MHz/v

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued	2					
4311	Frequency pushing figure	19	Ec and If = ±2% (max)				
	1/ DOD-023 and 2/ DOD-024						
	Injection electrode	-		---	---	0.2	MHz/v
	Heater	-		---	---	25	MHz/A
	3/ DOD-025						
	Injection electrode	-		---	---	0.3	MHz/v
	Heater	-		---	---	30	MHz/A
1367	Magnet polarity and strength	20		---	---	---	---
	Quality conformance inspection, part 3	2					
---	Life test	27	Group D: Po = 36 mW (min); Du = 100%; F1, F2, F3, F4, and F5	t	1,000	---	hrs
---	Life-test end points:						
4250	Power output	11	σ = 1.2:1 (worst phase)	Po	30	---	mW
4264	Ion noise modulation	13		Pn	82	---	dB
	Periodic-check tests						
---	Temperature, altitude and humidity	21	Operating	---	---	---	---
---	Vibration	22	Operating	---	---	---	---
---	Shock	23	No voltages	---	---	---	---
4258	RF interference shielding	24		---	---	1.0	mW
---	Acoustical noise	-	F = 100 to 20,000 Hz; reference level = 0.0002 dynes/cm^2; MIL-STD-810, method 515	---	---	130	dB

NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

2. The TUT shall be subjected to all quality conformance inspection tests which shall be performed in the sequence stated herein. These tests shall be measured at F1, F2, F3, F4, F5, and at room ambient conditions unless otherwise specified.

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NOTES: -Continued

3. The TUT shall be subjected to three cycles of the following test:

   (a) Lower temperature from room ambient to -80° F (-62° C)
   (b) Hold at -80° F for 1 hour
   (c) Raise from -80° to +185° F
   (d) Hold at +185° F for 1 hour
   (e) Lower temperature from +185° F to room ambient.

   The rate of change of temperature between hold periods shall not exceed 1.8° F/sec. The temperature changes specified in a, c, and e shall be accomplished in a period not to exceed 1 hour.

4. The TUT shall be vibrated in each of three mutually perpendicular axes, one of which is through the longest dimension for a period of 15 minutes.

5. The TUT shall be operated for a minimum period of 10 hours. Operation shall be at room ambient conditions and shall consist of power "on" for 40 minutes and all power "off" for 20 minutes. During power "on", the TUT shall operate over the specified frequency range. The time required to trace the frequency range shall not exceed the specified limits.

6. There shall be no frequency discontinuities in this frequency range.

7. The actual tuning curve shall be compared to the best straight line through the observed frequency versus voltage measurements. The deviation of any frequency on the actual tuning curve from the corresponding point on the best straight line expressed as a percentage of the center frequency of the tuning range shall not exceed the specified limit.

8. The tuning sensitivity is defined as the $\frac{\Delta F}{\Delta E_b}$ of the best straight line through the observed frequency versus voltage measurements. Slope deviation shall be defined as the deviation of slope of any 5 MHz portion of the actual tuning characteristic from the tuning sensitivity expressed as a percentage of the tuning sensitivity. ($F_2 - F_1 = 5$ MHz, corresponding to $\Delta E_b = E_{b2} - E_{b1}$; slope $= \frac{F_2 - F_1}{E_{b2} - E_{b1}} = \frac{5}{\Delta E_b}$. The slope deviation shall not exceed the specified limit.

9. The frequency versus voltage curve of each tube shall not deviate from the design value by more than the specified limits. The tuning sensitivity shall be as follows:

1/ DOD-023	1.00 MHz/v
2/ DOD-024	1.66 MHz/v
3/ DOD-025	2.88 MHz/v

10. The frequency deviation shall not be greater than the specified limits when operating into load VSWR of 2:1 which shall be varied through all phases to establish the maximum deviation.

11. The power output shall be continuous and not less than the specified limit at any frequency in the range F_1 through F_5.

12. The variation in power output into a load VSWR of 2:1 varied through all phases shall be less than the specified limits.

13. The noise output in the two 2 MHz sidebands at 60 ±4 MHz on both sides of the carrier shall be below the carrier (fundamental) by at least the specified limits.

14. The frequency versus voltage curve shall not deviate from its initial value by more than the specified limits when F1 thru F5 is traced in a period of 15 ±1 seconds and retraced in 2 ±0.5 seconds.

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NOTES: -Continued

15. The frequency change over the specified temperature range shall not exceed the specified limits.

16. The output peak to peak frequency modulation of the TUT shall not exceed the specified limits when subjected to vibration levels of 5 to 18 Hz, 0.12 DA.

17. The TUT shall be tuned to F1, F2, F3, F4, and F5 and the spurious power output shall be measured across the band. The suprious power output shall be below the fundamental frequency by at least the amount specified herein.

18. The value of the average ratio of frequency to anode voltage shall be as specified herein.

19. The change of frequency with heater current or injection voltage shall not exceed the specified limits.

20. Curves showing the effect of the magnetic interference caused by a uniformed magnetic field shall be supplied with each tube. The curves shall be specified for the three major axes and for both magnetic polarizations.

21. Temperature, altitude, and humidity. The TUT shall be placed in the test chamber in a manner similar to that in which it will be used in service. Sufficient connections shall be made to operate and monitor the performance during test. The TUT shall be exposed to the following conditions in the sequence shown:

Step	Temperature °F	Altitude (ft)	Duration
1	-80	0 (sea level)	1 hour
2	15	0	2 hours
3	165	75,000	1 hour
4	185	0	2 hours

Step 5 --- With the internal pressure of the chamber still at atmospheric pressure, the chamber temperature shall be adjusted to 14° F and stabilized. The test chamber door shall then be opened to allow frost to form on the equipment. The door shall be held open long enough for the frost to melt but not long enough to allow the moisture to evaporate. When the chamber door is opened, it is intended that frost will form, however, should the relative humidity of the air be such that frost will not form, an artificial means shall be used to provide the relative humidity necessary to have frost form. The chamber door shall be closed and cooling air at 35° F shall be supplied to the equipment. With voltages applied, the tube in the chamber shall meet the requirements of this specification. Data shall be obtained and compared to the data obtained prior to starting this procedure.

The rate of change of temperature between steps shall not exceed 1.8° F/sec. The change from room ambient to -80° F shall not exceed 4 hours. The power output of the TUT shall be measured and examined at each step for "holes" throughout the operating frequency ranges of F1, F2, F3, F4, and F5. The measurements of frequency and power at the different steps of the environmental exposure shall be performed at the same five frequencies. After completion of the test, the tube shall be tested and shall meet the requirements of quality conformance inspection, parts 1, 2, and 3.

22. The TUT shall be tested in accordance with MIL-STD-810, method 514. The TUT shall be operated at F3 throughout the vibration test with the output power being monitored. Changes in frequency and output power occurring during this test shall be recorded together with the conditions occurring at the time, (i.e., resonance, G level, vibrating frequency, etc.). Subsequent to this test, the TUT shall exhibit no mechanical damage and shall satisfy the requirements of quality conformance inspection, parts 1, 2, and 3.

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NOTES: -Continued

23. The TUT shall be tested in accordance with MIL-STD-810, method 516. After completion of this test, the TUT shall exhibit no mechanical damage and shall satisfy the requirements of quality conformance inspection, parts 1, 2, and 3.

24. This test shall be conducted with the TUT operating in a swept condition at 60 Hz rate over F1 thru F5. Measurements shall be made using a Polarad calibrated microwave test antenna Model CA-S (L), or equivalent coupled to the TUT. The coupling shall be adjusted by proper antenna orientation to obtain maximum response. The antenna shall be connected directly to a bolometer and a Hewlett-Packard Model 431 power bridge, or equivalent. The radiated interference generated by the TUT shall not exceed the specified limits.

25. Care shall be exercised to prevent demagnetization when handling and mounting the tube. Ferromagnetic materials or energized magnets shall not be brought within 2 inches of the tube.

26. The values of the parameters specified in quality conformance inspection, parts 1, 2, and 3 shall first be measured and recorded. The operating (test) conditions for each test, i.e., electrode voltages, currents, etc., shall be recorded. Where the value of a parameter is required over the operational frequency range, the number of frequency points at which measurements are made shall be a minimum of F1 through F5. A copy of the critical performance characteristics including operating voltages shall be shipped with each tube. The operating voltages (table 1 of figure 1) shall be clearly indicated on any exposed surface of the tube. Each tube delivered on the contract shall operate optimally under a single combination of discrete voltages which fall within the ranges shown under absolute ratings. All operating requirements of this specification shall be met under these specific voltages.

27. The operating frequency shall be changed after each 200-hour operating period.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11
DSA - ES

User activities:
Army - MU, WC
Navy - AS, OS, MC, CG, SH
Air Force - 17

Preparing activity: Navy - EC

Agent: DSA - ES

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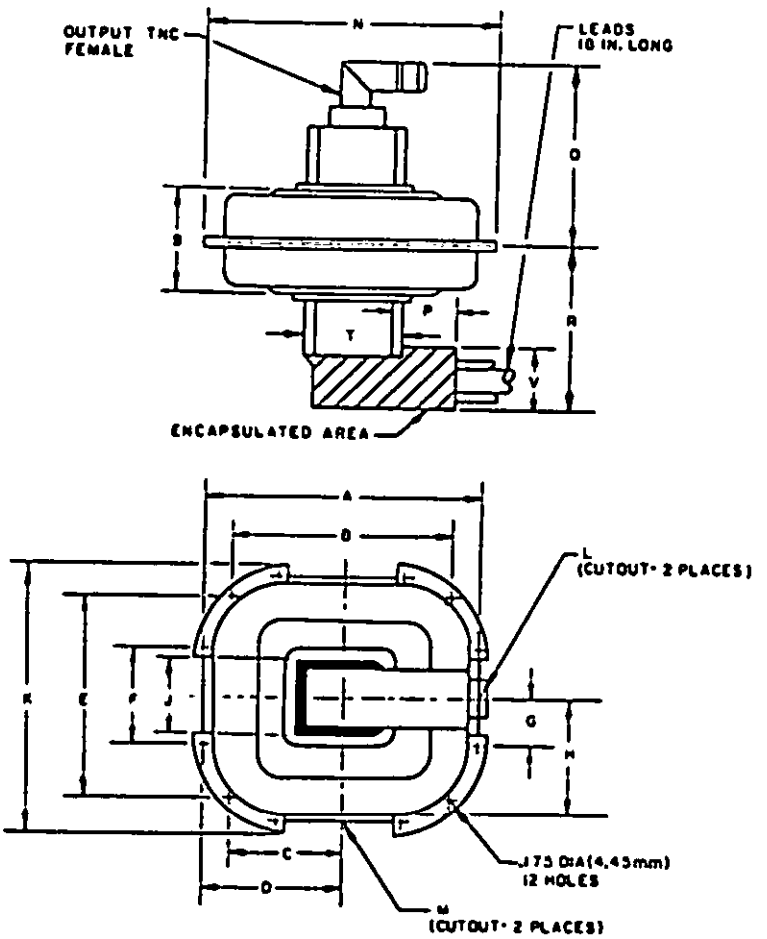


FIGURE 1. Outline drawing of electron tubes 1/DOD- 023 , 2/DOD-024 , and 3/DOD- 025 .

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SCHEMATIC DIAGRAM

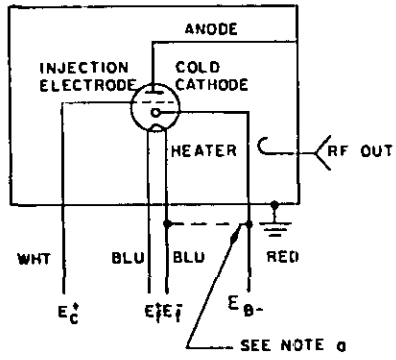


TABLE 1. Marking

Manufacturer name and code	
Tube type	
Serial No.	
Wire code	
Lead	Element
Wht	Injection electrode
Blu	Heater (See note c)
Blu	Heater
Red	Cold cathode
Case	Anode

LTR	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
	Quality conformance inspection, part 1 (see note 1)	
A	.5.330 (135.38)	5.350 (135.89)
B	4.030 (102.36)	4.050 (102.87)
C	2.010 (51.05)	2.030 (51.56)
D	2.660 (67.56)	2.680 (68.07)
E	3.450 (87.63)	3.470 (88.14)
F	2.150 (54.61)	2.170 (55.12)
G	1.070 (27.18)	1.090 (27.69)
H	1.720 (43.69)	1.740 (44.20)
J		.800 (20.32)
K		4.325 (109.86)
L	1.594x.250 (40.49x6.35)	1.656x.312 (42.06x7.92)
M	1.594x.281 (40.49x7.14)	1.656x.343 (42.06x8.71)
N		6.031 (153.19)
O		2.968 (75.39)
P		.375 (9.53)
R		2.000 (50.80)
S		2.150 (54.61)
T		1.400 (35.56)
V		.375 (9.53)

NOTES:
a. An external connection is required between the cold cathode lead and one of the heater leads.
b. The information in table I shall be marked on the tube on any exposed surface. Characters shall be permanently marked in white .125 inch high with a non-fungus nutrient material.
c. Mark one heater lead with heat shrinkable sleeve. The sleeve shall be white with "Ef" printed in black. The sleeve shall be .500 ± .125 inch long, place 2.00 ± .25 inches from the boot.
d. Finish shall be Strathmore X-2552 black enamel in accordance with TT-E-489.

FIGURE 1. Outline drawing of electron tube types 1/DOD-023, 2/DOD-024, and 3/DOD-025 - Continued.

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